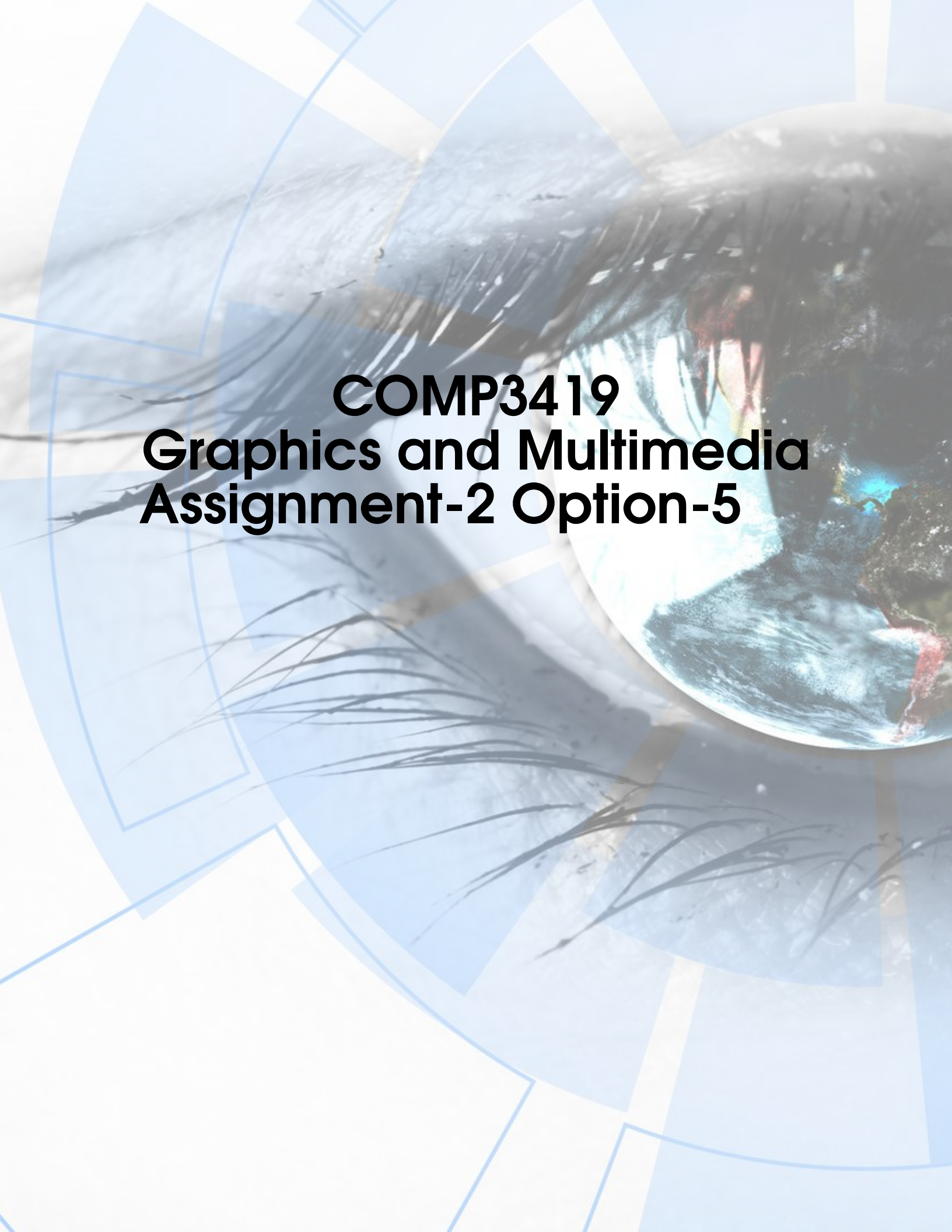
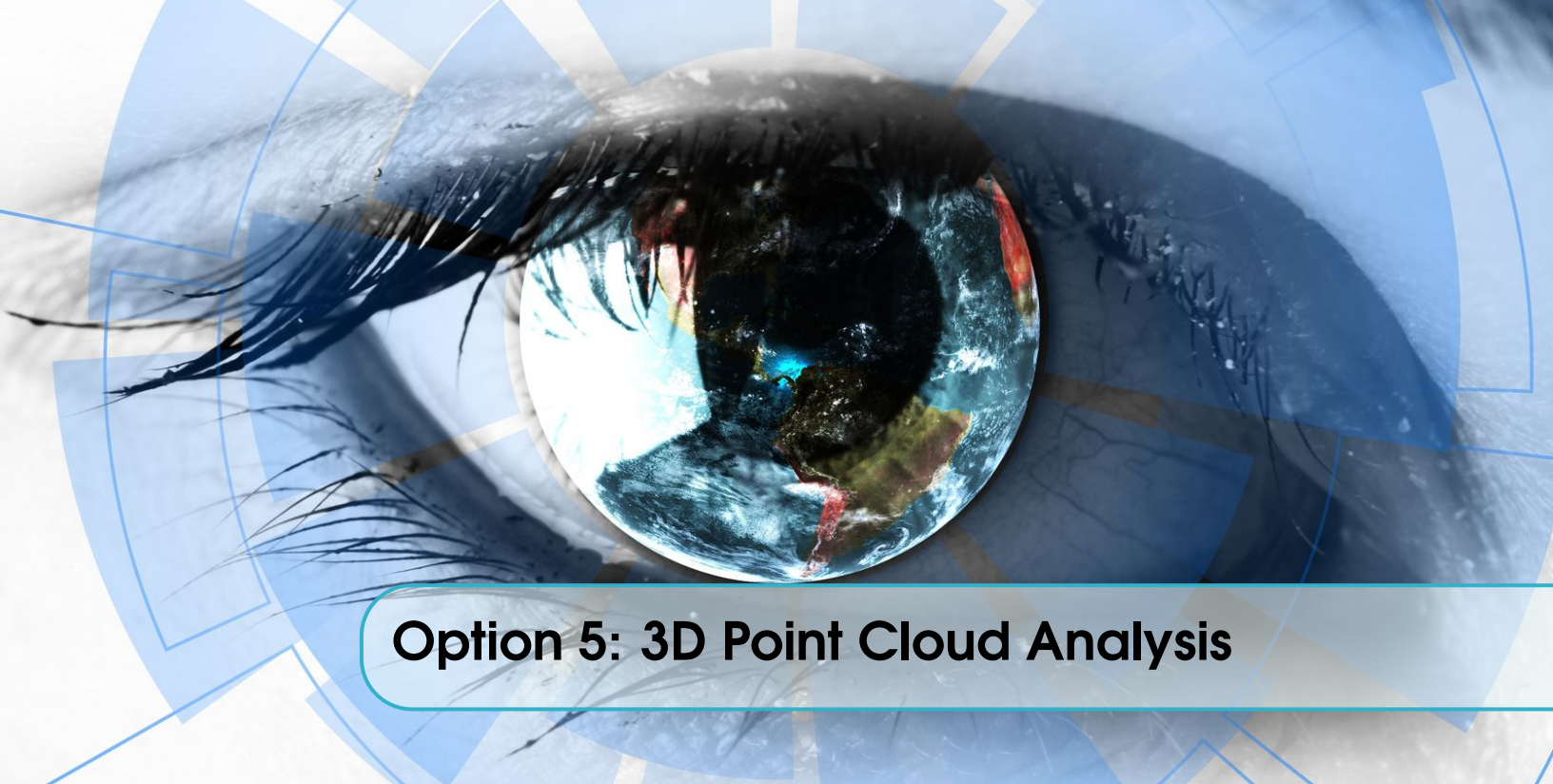


The background is a complex digital composition. It features a large, semi-transparent eye in the center, with a detailed iris and pupil. The eye is overlaid on a globe of the Earth, showing continents and oceans. The entire scene is framed by a series of overlapping, semi-transparent blue and white geometric shapes, including circles and lines, creating a layered, futuristic aesthetic.

COMP3419

Graphics and Multimedia

Assignment-2 Option-5



Option 5: 3D Point Cloud Analysis

1 Key Information

- The mark of "**COMP3419 Assignment-2 Multimedia Project (Option-3: 3D Point Cloud Analysis)**" will be given on canvas submission. Due time: before 23:59, Saturday of Week 13 (09-Nov-2025).
- This individual assignment is worth **16%** of your final assessment.
- **ModelNet40** dataset can be downloaded from [here \(1.6GB\)](#).
- **Submission Deliverables:** Students are asked to create a **zip** file of all deliverables, including a project report (written in $\text{\LaTeX}$), all source code (with your best models saved), and a demo video. Please be aware of the following submission **restrictions** that (1) this zip file should be named as "SIDxxx_Ass2Opt3.zip" where xxx denotes the student ID (e.g., "SID450003419_Ass2Opt3"), and (2) the demo video should be named as "SIDxxx_Ass2Opt3.mp4" as well. Failing to follow these restrictions or missing of demo video would cause a 5-mark penalty.
- Students' assignments **will only be marked if** all deliverables can be **accessed** from the Canvas System, and they can be **runnable** following instructions provided in README txt file. Once plagiarism detected by the Canvas system, the student will receive no mark immediately, as well as other related penalties from university.
- A **README** txt file (to describe the steps/instructions regarding how to get the source code running to derive the expected outputs) would help markers get familiar with the submission.

2 General Marking Policy

Late Submission Policy

For the late submission cases, penalties will be assigned according to the university wide late penalties for assignment Clause 7A of the Assessment Procedures.

Special Consideration and Arrangements

While you are studying, there may be circumstances or essential commitments that impact your academic performance. Our special consideration and special arrangements process is there to

support you in these situations. More information on how to lodge the special consideration application, can be found from this [webpage](#).

3 Introduction

The aim of this individual assignment (Assignment-2 Option-3) is to perform literature reviews, reproduce and evaluate methods for 3D point cloud classification across different object categories. Here we introduce a public large-scale 3D model dataset - ModelNet.

4 Dataset Description

- The specific dataset used in this assignment is ModelNet40, containing 12,311 pre-aligned 3D shapes in the form of point clouds. The overall data come from 40 different categories including airplane, chair, sofa, etc. These data has been split into 9,843 (80%) for training and 2,468 (20%) for testing.
- The class types can be found in **modelnet40_shape_names.txt**. File names of training data are contained in **modelnet40_train.txt**, and those of testing data are included in **modelnet40_test.txt**. The main objective is a multi-label classification task to distinguish the type of the examined 3D object.
- Both the training and testing data (*.txt files) contain 10000 points with the corresponding XYZ positions (1st - 3rd columns) and XYZ normal vectors (4th - 6th columns).
- Please note that you are **only required** to use 1024 out of 10000 points, with the point XYZ coordinates for this task.
- The model performance should be evaluated by the classification accuracy.

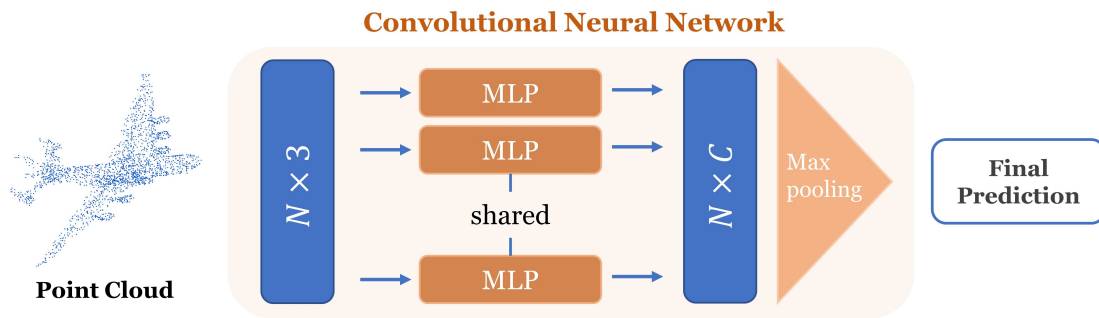


Figure 1: Deep learning model for 3D point cloud classification, where a lightweight architecture of PointNet is shown. N and C denote the number of points and category classes, respectively.

5 Literature review and reproduction (8%)

- 6% Students are requested to preform **three** literature reviews on existing point-based works. Please note that you can only select at most two approaches from PointNet, PointNet++ and DGCNN. For each chosen work, you have to reproduce and display the classification result in the project report (1%), and discuss and compare the advantages and disadvantages among them (1%).
- 2% Students are requested to shows their attempts in method training and fine-tuning their model results. You should also complete performance comparison (following the official dataset split), and discuss the significance of their selected approaches.

 You are feel free to explore any existing approach proposed in all online publicly-available academic papers, and remember to reference them correctly and appropriately in your report.

6 Project Report (8%)

The report should contain introduction, literature review, experimental setup, results & discussion, conclusion, and references. It should be 2 – 6 pages (maximum 8) in two-column layout, using this template (<http://www.latextemplates.com/template/wenneker-article>) to follow the scientific style and formatted with $\text{\LaTeX}$ ¹. A brief guideline is as follows:

- 1% Introduction: introducing the project aim, methods and findings (suggested in 200 - 300 words; no more than 500 words).
- 2% Literature review: individual review of the three selected approaches and their implementation details.
- 1% Experimental setup: describing the dataset and evaluation metrics (suggested in 100 - 200 words; no more than 250 words).
- 2% Results & Discussion: presenting the evaluation results of each method, the advantage and disadvantage of the selected approach. Based on the experiment results, insightful discussion is expected to demonstrate the corresponding analysis performed. Tables and figures are preferred to be used for result demonstrations.
- 1% Conclusion: summarising the study and findings (suggested in 100 - 120 words; no more than 150 words).
- 1% References: listing literature and other references (papers and/or online resources).

7 Deliverables

Students are requested to create a zip file of all deliverables, including:

- All the related source code (with all three models saved).
- A PDF project report formatted with $\text{\LaTeX}$.
- A demo video (recorded in Zoom) to introduce your literature reviews, experimental results and findings.
- See *Key Information* section for submission restrictions.

¹Do not worry if you have not used latex before. It has a similar syntax to HTML. You can find a handy online latex editor at <https://www.sharelatex.com/>.

- R Demo Video:** The presentation of your literature reviews, findings, as well as your final results, should be recorded as a video, using **Zoom** for its screen-sharing function, and the expected duration is 3-5 mins (no more than 5 mins). It is suggested to use Microsoft PowerPoint/Google Slides along with the experiment results to present your model performance, following the same structure of the project report.

8 Bonus Section: 4 Marks

The aim of this section is for advanced point cloud processing, where 2 challenging tracks are provided for you to complete. Please note that if you try to attempt the bonus section, you should add a separate "Bonus" section in your report (before "conclusion" section). Friendly reminder: **you only need to complete one track** among the two provided below, and each track is worth 4 marks. But the full mark of your assignment 2 would still be capped at 16 marks.

8.1 Track 1: 3D Part Segmentation

The objective of segmentation is to build computational techniques that predict the label of each point within the object. Here we introduce a public large-scale dataset - ShapeNetPart. You can download the dataset [here \(674MB\)](#).

Dataset Description

- ShapeNetPart includes 14,006 training samples and 2,874 testing samples from 16 object classes with 50 part labels. Totally, there are 16 objects including airplane, bag, cap, car, etc. For each object, the fine-grained point labels are provided accordingly.
- Both the training, validation and testing data (*.txt files) contain over 2,000 points with the corresponding XYZ positions (1st - 3rd columns), XYZ normal vectors (4th - 6th columns), and the point label (7th column).
- Files named by numbers represent different objects (please refer to **synsetoffset2category.txt**). File names of training data are contained in **train_test_split/shuffled_train_file_list.json**, file names of validation data are contained in **train_test_split/shuffled_val_file_list.json**, and those of testing data are included in **train_test_split/shuffled_test_file_list.json**.
- Please note that you are **only required** to use 2048 points with XYZ coordinates and point labels for this task.

Evaluation Metric

In this task, we use mean class Intersection over Union (mIoU) for performance evaluation. Assume that there are a total of C classes, and p_{ij} represents the predicted label of a point belongs to i while the true label is j . In other words, p_{ii} represents the true positive, while p_{ij} and p_{ji} represent false positives and false negatives respectively. The mIoU computes the intersection ratio between ground truth and predicted value averaged over the total number of classes C as follows:

$$mIoU = \frac{1}{C} \sum_{i=1}^C \frac{p_{ii}}{(\sum_{j=1}^C p_{ij}) + (\sum_{j=1}^C p_{ji}) - p_{ii}}.$$

Task

- 1% Literature review: you should review the model behavior of the three selected methods on the segmentation task.

- 1% Quantitative Results: you should provide the final mIoU value of your chosen method.
- 1% Qualitative Results: you should provide at least 5 visualized results of the segmented outputs based on your chosen approach, regardless of the object class. Please note that you can refer to [this link](#) for your output visualization.
- 1% Discussion: you should compare the advantage and disadvantage of your chosen approaches.

8.2 Track 2: Exploration

The aim of this track is to motivate students to explore some latest techniques that might be beneficial for 3D point cloud classification task. Students are encourage to try out cutting-edge computer vision and pattern recognition techniques to enhance **one** of their selected baseline works, by using some recent approaches such as transformers, attention modules, self-supervision and graph neural network, etc.

Task

- 1% Introduction: introducing your interested technique and idea about how you would modify the existing model.
- 1% Literature review: presenting a brief review on some related works from which you get your idea.
- 1% Explanation: explaining clearly about how you adopt the technique for the classification task.
- 1% Results and Discussion: reporting and discussing your new result based on your enhanced model design.

9 Google Cloud Platform Education Grant for COMP3419 Students

You can either work locally or use the remote computing platform [Google Cloud Engine](#) (click here), which is funded by Google Cloud Platform Education Grant for COMP3419 students, to gain a better computational resources (CPU, GPU, TPU, etc).

Student Google Cloud Coupon Retrieval Access

- The coupon code exclusively for COMP3419 can be redeemed [here](#). Please keep in mind that there is a redemption limit of 5 coupons per account.
- You will be asked to provide your name and school email address (e.g., unikey@uni.sydney.edu.au), which needs to match the domain (**please choose the default domain@uni.sydney.edu.au**). An email will be sent to you to confirm these details before a coupon is sent to you.
- Watch the Google Cloud Platform Essentials video series on YouTube.
- Instructions for [how to redeem grants](#) in Google Cloud Platform and click "**Redeem Now**" button.
- A **comprehensive tutorial** on Google Cloud setup and dependencies setup for deep learning-based AI projects can be found from [here](#).
- Please **ensure your instances closed** every time when you finish using or training in Google Cloud, to avoid unexpected waste in your coupon.